Motion, forces and energy

Mark scheme and worked solutions for 90 questions: 45 multiple choice, 45 structured.

1 · Part 1

Core · Physical quantities and measurement techniques · [1]

Answer: B

Worked solution

When an object does not start at the zero mark, subtract the start reading from the end reading. 15.5 − 2.0 = 13.5 cm.

2 · Part 1

Core · Physical quantities and measurement techniques · [1]

Answer: D

Worked solution

Timing many oscillations and dividing by the number of oscillations gives a more accurate period than timing one. 31.6 / 20 = 1.58 s.

3 · Part 1

Core · Physical quantities and measurement techniques · [1]

Answer: A

Worked solution

The water displaced by the ball bearings is equal to their total volume. Find the rise in level, then divide by the number of ball bearings: 4.8 cm^3 / 40 = 0.12 cm^3. Measuring many small objects together and dividing makes the result more accurate.

4 · Part 1

Extended · Physical quantities and measurement techniques · [1]

Answer: C

Explanation

Momentum, weight and gravitational field strength all have magnitude and direction, so they are all vectors. A contains mass, B contains speed and D contains distance and energy, which are scalars.

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5 · Part 1 Extended Physical quantities and measurement techniques [1]

Answer: A

Explanation

Both cars have the same speed, 15 m / s. Velocity is speed in a given direction, and the directions differ (north and east), so the velocities are different. B swaps the ideas of speed and velocity. C ignores direction. D is wrong because the speeds are equal.

6 · Part 1 Extended Physical quantities and measurement techniques [1]

Answer: B

Worked solution

Draw the two forces head to tail: they form the two shorter sides of a right-angled triangle, and the resultant is the longest side. By Pythagoras, resultant^2 = 7.0^2 + 24^2 = 625, so the resultant is 25 N.

7 · Part 1 Core Motion [1]

Answer: D

Worked solution

speed = distance / time. To get m / s, the distance must be in metres and the time in seconds. 4500 m / 900 s = 5.0 m / s. Check: 18 km / h ÷ 3.6 = 5.0 m / s.

8 · Part 1 Core Motion [1]

Answer: C

Worked solution

Average speed is never found by averaging the speeds unless the times are equal. Work out each time, add them, then divide the total distance by the total time: 24 km / 0.50 h = 48 km / h.

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9 · Part 1 Core Motion [1]

Answer: B

Explanation

From 0 to 4.0 s the robot travels 6.0 m in each 2.0 s, so its speed is constant. From 4.0 s to 8.0 s the distance does not change, so it is at rest. A and D are wrong because equal distances in equal times show constant speed, not acceleration. C ignores the readings after 4.0 s.

10 · Part 1 Core Motion [1]

Answer: A

Explanation

Near the Earth's surface, with negligible air resistance, the acceleration of free fall is approximately constant at about 9.8 m / s^2. B confuses increasing speed with increasing acceleration. C gives the wrong unit (m / s is a unit of speed). D would only happen if air resistance were significant.

11 · Part 1 Extended Motion [1]

Answer: D

Worked solution

Change in velocity = final − initial = 9.0 − 24 = −15 m / s. Divide by the time: −15 / 30 = −0.50 m / s^2. A deceleration is a negative acceleration.

12 · Part 1 Extended Motion [1]

Answer: C

Worked solution

Find the change in speed in each 1.0 s interval. If the changes are equal, the acceleration is constant; if they differ, the acceleration is changing. Only C has unequal changes.

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13 · Part 1 Extended Motion [1]

Answer: A

Explanation

At terminal velocity the speed is constant, so the acceleration is zero; this happens when the upward air resistance balances the weight. B describes falling with no air resistance. C would make the raindrop slow down, not fall at a steady speed. D describes the raindrop before it reaches terminal velocity.

14 · Part 1 Core Motion [1]

Answer: B

Worked solution

The gradient of a straight-line section must use the changes between two points on that section: Δdistance / Δtime = 12.0 m / 4.0 s = 3.0 m / s.

15 · Part 1 Core Motion [1]

Answer: C

Worked solution

Split the journey into stages. While the speed increases steadily from zero, the distance is the area of a triangle, ½ × base × height. At constant speed the distance is speed × time. Add the stages: 24 m + 30 m = 54 m.

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16 · Part 1 Core Physical quantities and measurement techniques [4]

Mark scheme

(a) **[1]**

- one card is too thin to measure accurately with a ruler / measuring many cards together makes the (percentage) error in the result smaller (1)

(b) **[1]**

- 2.6 − 1.0 = 1.6 cm (1)

ALLOW 16 mm

(c) **[2]**

- thickness = 1.6 / 50 = 0.032 cm (1)
- = 0.32 mm (1)

ECF from the height in (b)

REJECT final answer 0.032 mm or 3.2 mm (unit conversion error), award the first mark only

Worked solution

(a) A single card is far thinner than the smallest division on a ruler (1 mm), so measuring it alone would give a very inaccurate result. Measuring 50 cards and dividing by 50 spreads the reading uncertainty over all the cards.

(b) The stack starts at the 1.0 cm mark, not at zero, so subtract: 2.6 cm − 1.0 cm = 1.6 cm.

(c) One card = 1.6 cm / 50 = 0.032 cm. There are 10 mm in 1 cm, so 0.032 cm × 10 = 0.32 mm. Check: 0.32 mm × 50 = 16 mm = 1.6 cm.

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17 · Part 1 Core Physical quantities and measurement techniques [6]

Mark scheme

(a) **[1]**

- one oscillation takes too short a time to measure accurately / the reaction time is a large fraction of one oscillation, so timing 10 makes the (percentage) error smaller (1)

(b) **[2]**

- average time for 10 oscillations = (16.4 + 16.8 + 16.6) / 3 = 16.6 s (1)
- period = 16.6 / 10 = 1.66 s (1)

ALLOW 1.7 s

ECF from an incorrect average divided by 10

(c) **[1]**

- any one from: (1)
- use a fixed marker (e.g. a pin or line) at the centre / lowest point of the swing and start and stop timing as the mass passes it
- start the stopwatch as the mass passes a chosen point and count "zero" at that moment, not "one"

ALLOW time more oscillations (e.g. 20)

IGNORE "be more careful", "repeat"

(d) **[2]**

- human reaction time (about 0.2 s or more) is similar to or larger than the 0.1 s being measured, so the error would be very large (1)
- use a light gate (or two light gates) connected to a digital timer / data logger, which starts and stops automatically (1)

Worked solution

(a) A stopwatch is started and stopped by hand, so each reading has an error caused by reaction time. Over 10 oscillations this error is the same size but is a much smaller fraction of the total time.

(b) First average the three repeat readings: 49.8 s / 3 = 16.6 s. This is for 10 oscillations, so one period = 16.6 / 10 = 1.66 s. Check: 1.66 s × 10 × 3 = 49.8 s.

(c) The mass moves fastest at the lowest point of its swing, so it passes a fixed marker there at a clearly defined moment. This makes it easier to judge exactly when to start and stop.

(d) A time of 0.1 s is shorter than a typical human reaction time, so a hand-operated stopwatch cannot measure it reliably. A light gate detects the car interrupting a beam and starts and stops a digital timer automatically, with no reaction time.

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18 · Part 1 Extended Physical quantities and measurement techniques [4]

Mark scheme

(a) **[1]**

- a scalar has magnitude (size) only; a vector has magnitude and direction (1)

(b) **[2]**

- vectors: momentum, electric field strength, acceleration (1)
- scalars: distance, temperature, speed (1)
- each mark needs all three quantities correct and no extras

(c) **[1]**

- weight is a force, which acts in a direction (downwards / towards the centre of the Earth), so it is a vector (1)

> **IGNORE** references to how bathroom scales work

Worked solution

(a) A scalar is described by a size alone. A vector needs a size and a direction.

(b) Momentum, electric field strength and acceleration all have a direction, so they are vectors. Distance, temperature and speed have size only, so they are scalars.

(c) The way a quantity is measured does not decide whether it is a scalar or a vector. Mass is the amount of matter and has no direction. Weight is the gravitational force on an object and always acts in a direction, towards the centre of the Earth, so it is a vector.

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19 · Part 1 Extended Physical quantities and measurement techniques [4]

Mark scheme

(a) **[2]**

- resultant^2 = 72.0^2 + 135^2 OR resultant = √(5184 + 18225) (1)
- 153 N (1)

ALLOW scale-drawing answer in the range 150 N to 156 N

(b) **[2]**

- tan θ = 135 / 72.0 (1)

ALLOW sin θ = 135 / 153 or cos θ = 72.0 / 153 (ECF from (a))

- θ = 61.9° (1)

ALLOW 62°; scale-drawing answer in the range 60° to 64°

REJECT 28.1° (this is the angle with the 135 N force)

Worked solution

(a) The two forces are at right angles, so they form the two shorter sides of a right-angled triangle and the resultant is the hypotenuse. resultant = √(72.0^2 + 135^2) = √23 409 = 153 N. You cannot just add 72.0 N and 135 N, because the forces act in different directions.

(b) In the triangle, the 135 N side is opposite the angle θ and the 72.0 N side is adjacent to it. tan θ = 135 / 72.0 = 1.875, so θ = 61.9°. Check: cos 61.9° × 153 ≈ 72.0 N. The resultant points north of east, at 61.9° to the east direction.

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20 · Part 1 Extended Physical quantities and measurement techniques [6]

Mark scheme

(a) **[2]**

- wind speed^2 = 13^2 − 12^2 OR wind speed = √(169 − 144) (1)
- 5.0 m / s (1)

(b) **[2]**

- tan θ = 5.0 / 12 OR sin θ = 5.0 / 13 OR cos θ = 12 / 13 (1)
- θ = 23° (1)

ALLOW 22.6°

ECF from the wind speed in (a)

(c) **[2]**

- velocities are vectors, so their directions must be taken into account (1)
- the wind is at right angles to the drone's velocity, so they combine using Pythagoras / a vector triangle; values only add directly when they act in the same direction (1)

Worked solution

(a) The drone's velocity relative to the air (12 m / s north) and the wind velocity (east) are at right angles. The resultant, 13 m / s, is the hypotenuse of the right-angled triangle. So wind speed = √(13^2 − 12^2) = √25 = 5.0 m / s. Check: √(12^2 + 5.0^2) = √169 = 13 m / s.

(b) The wind side (5.0 m / s) is opposite the angle from north, and the 12 m / s side is adjacent to it. tan θ = 5.0 / 12 = 0.417, so θ = 23°. The drone's actual path is 23° east of north.

(c) Adding 12 m / s and 5.0 m / s to get 17 m / s is only correct if the wind blows in the same direction as the drone flies. Here the two velocities are at right angles, so the resultant must be found with Pythagoras, giving 13 m / s, not 17 m / s.

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21 · Part 1 Core Motion [6]

Mark scheme

(a) **[1]**

- speed is the distance travelled per unit time (1)

> **ALLOW** distance travelled divided by time taken

(b) **[1]**

- velocity is speed in a given direction / velocity has direction as well as size but speed does not (1)

(c) **[3]**

- total distance = 18 + 27 = 45 km (1)
- total time = 20 + 10 + 30 = 60 min = 1.0 h (1)
- average speed = 45 / 1.0 = 45 km / h (1)

> **ECF** from an incorrect total distance or total time (e.g. 54 km / h if the 10 min stop is left out loses the second mark only)

(d) **[1]**

- the van travels no distance during the 10 min stop, but this time is still included in the total time (1)

Worked solution

(a) Speed is the distance travelled per unit time.

(b) Velocity is speed in a given direction, so two objects with the same speed can have different velocities if they move in different directions.

(c) Average speed = total distance travelled / total time taken. Total distance = 18 km + 27 km = 45 km. The total time must include the stop: 20 + 10 + 30 = 60 min = 1.0 h. Average speed = 45 km / 1.0 h = 45 km / h. Check: 45 km / h for 1.0 h gives 45 km.

(d) While the van is stopped, time passes but no distance is covered. Including the stop in the total time lowers the average speed below the speed of either moving stage.

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22 · Part 1 Core Motion [7]

Mark scheme

(a) **[2]**

- she is at rest / stationary (1)
- the distance stays at 90 m / does not change between 30 s and 50 s (1)

(b) **[2]**

- speed = 90 / 30 OR gradient = Δdistance / Δtime (1)
- 3.0 m / s (1)

(c) **[1]**

- (180 − 90) / (70 − 50) = 4.5 m / s (1)

(d) **[2]**

- average speed = 180 / 70 (1)
- 2.6 m / s (1)

ALLOW 2.57 m / s

REJECT mean of the stage speeds

Worked solution

(a) Between 30 s and 50 s the distance stays at 90 m, so she is not moving: she is at rest. On a distance–time graph this is a horizontal line.

(b) The speed is the gradient of the straight-line section: 90 m / 30 s = 3.0 m / s. Check: 3.0 m / s × 10 s = 30 m in each 10 s interval, matching the table.

(c) Between 50 s and 70 s she travels 180 − 90 = 90 m in 20 s, so her speed is 90 / 20 = 4.5 m / s.

(d) Average speed = total distance / total time = 180 m / 70 s = 2.57 m / s, which rounds to 2.6 m / s. The rest time is included, which is why this is lower than either running speed.

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23 · Part 1 Core Motion [7]

Mark scheme

(a) **[2]**

- first stage: accelerating / speed increasing (at a constant rate); second stage: moving at constant speed; third stage: decelerating / speed decreasing (at a constant rate) to rest
- all three correct (2); two correct (1)

(b) **[2]**

- distance = area under graph = ½ × 4.0 × 2.5 (1)
- 5.0 m (1)

(c) **[3]**

- constant-speed stage: 2.5 × 12.0 = 30 m (1)
- final stage: ½ × 5.0 × 2.5 = 6.25 m (1)
- total = 5.0 + 30 + 6.25 = 41 m (1)

ALLOW 41.25 m or 41.3 m

ECF from (b)

Worked solution

(a) The first line slopes upwards, so the speed increases steadily: the lift accelerates. The second line is horizontal, so the speed is constant at 2.5 m / s. The third line slopes down to zero, so the lift decelerates steadily and stops.

(b) The distance travelled is the area under the speed–time graph. For the first stage this is a triangle: ½ × 4.0 s × 2.5 m / s = 5.0 m.

(c) The second stage is a rectangle lasting 16.0 − 4.0 = 12.0 s: 2.5 × 12.0 = 30 m. The third stage is a triangle lasting 21.0 − 16.0 = 5.0 s: ½ × 5.0 × 2.5 = 6.25 m. Total = 5.0 + 30 + 6.25 = 41.25 m, which is 41 m to 2 significant figures. Check with the trapezium shape: ½ × (21.0 + 12.0) × 2.5 = 41.25 m.

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24 · Part 1 Core Motion [5]

Mark scheme

(a) **[2]**

- accelerating (1)
- the speed increases with time / increases by the same amount (0.98 m / s) in each 0.10 s (1)

(b) **[1]**

- 9.8 m / s^2 (1)

REJECT wrong unit, e.g. 9.8 m / s

(c) **[2]**

- distance = area under graph = ½ × 0.40 × 3.92 (1)
- 0.78 m (1)

ALLOW 0.784 m

Worked solution

(a) The speed rises from 0 to 3.92 m / s, so the ball is accelerating. It gains the same speed, 0.98 m / s, in every 0.10 s, so it speeds up steadily.

(b) Near the Earth's surface the acceleration of free fall is approximately constant, about 9.8 m / s^2.

(c) The distance fallen is the area under the speed–time graph. The graph is a straight line from the origin, so the area is a triangle: ½ × 0.40 s × 3.92 m / s = 0.784 m, about 0.78 m. Check: the average speed is 3.92 / 2 = 1.96 m / s, and 1.96 × 0.40 = 0.784 m.

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25 · Part 1 Extended Motion [6]

Mark scheme

(a) **[1]**

- acceleration is the change in velocity per unit time (1)

(b) **[2]**

- $a = \Delta v / \Delta t = (8.0 - 2.0) / 6.0$ (1)
- 1.0 m / s^2 (1)

(c) **[2]**

- $a = (0 - 8.0) / (15.0 - 9.0)$ (1)
- −1.3 m / s^2 (1)

ALLOW −1.33 m / s^2

- the negative sign is needed for the second mark

(d) **[1]**

- 1.3 m / s^2 (1)

ALLOW 1.33 m / s^2

ECF the size of the answer to (c)

Worked solution

(a) Acceleration is the change in velocity per unit time.

(b) Between 0 and 6.0 s the speed goes from 2.0 m / s to 8.0 m / s. $a = \Delta v / \Delta t = 6.0 / 6.0 = 1.0$ m / s^2. This is the gradient of the first section of the speed–time graph. Check: at 3.0 s the speed is $2.0 + 1.0 \times 3.0 = 5.0$ m / s, matching the table.

(c) Between 9.0 s and 15.0 s the speed goes from 8.0 m / s to 0. $a = (0 - 8.0) / 6.0 = -1.33$ m / s^2, which is −1.3 m / s^2. The negative value shows she is slowing down. Check: at 12.0 s the speed is $8.0 - 1.33 \times 3.0 = 4.0$ m / s.

(d) A deceleration is a negative acceleration, so the deceleration is the size of the answer to (c): 1.3 m / s^2.

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26 · Part 1 Extended Motion [6]

Mark scheme

(a) **[2]**

- changing acceleration (1)
- the speed increases by different amounts in each 1.0 s interval (6.0, 5.0, 4.0, 3.0, 2.0 m / s), so the acceleration decreases (1)

(b) **[2]**

- $a = \Delta v / \Delta t = (20.0 - 0) / 5.0$ (1)
- 4.0 m / s^2 (1)

(c) **[2]**

- between t = 0 and t = 1.0 s (1)
- 6.0 m / s^2 (1)

Worked solution

(a) Work out the change in speed in each second: 6.0, 5.0, 4.0, 3.0 and 2.0 m / s. These are not equal, so the acceleration is changing. Each change is smaller than the one before, so the acceleration is decreasing even though the car is still speeding up.

(b) Average acceleration = total change in velocity / total time = 20.0 m / s / 5.0 s = 4.0 m / s^2.

(c) The biggest change in speed in a single second is 6.0 m / s, between 0 and 1.0 s. So the average acceleration in that interval is 6.0 / 1.0 = 6.0 m / s^2. Check: it is larger than the 4.0 m / s^2 average over 5.0 s, as it should be when the acceleration decreases over time.

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27 · Part 1 Extended Motion [5]

Mark scheme

(a) **[2]**

- they fall together / reach the ground at the same time (1)
- there is no air resistance, so both fall with the same acceleration (the acceleration of free fall) (1)

(b) **[2]**

- on the Earth the feather falls more slowly / the hammer reaches the ground first (1)
- air resistance on the feather is large compared with its weight, so its acceleration is much smaller (the feather quickly reaches a low terminal velocity) (1)

(c) **[1]**

- the constant (maximum) velocity reached by a falling object when the air or liquid resistance is equal to its weight / when the resultant force is zero (1)

Worked solution

(a) With no atmosphere there is no air resistance. The only force on each object is its weight, and in a uniform gravitational field every object falls with the same acceleration whatever its mass. So the hammer and feather speed up together and land at the same moment.

(b) In air, both objects have air resistance acting upwards. For the hammer this force is tiny compared with its weight, so it falls almost as it would without air. For the feather the air resistance soon becomes as large as its weight, so it stops accelerating and drifts down slowly. The hammer lands first.

(c) As a falling object speeds up, the air resistance on it increases. When the air resistance equals the weight, the object stops accelerating and falls at a constant speed: its terminal velocity.

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28 · Part 1 Extended Motion [6]

Mark scheme

(a) **[2]**

- top speed = acceleration × time OR $\Delta v = a \times \Delta t = 0.60 \times 40$ (1)
- 24 m / s (1)

(b) **[2]**

- time = $\Delta v / a = 24 / 0.80$ (1)
- 30 s (1)

ECF from (a)

(c) **[2]**

- distance = area under speed–time graph = ½ × 30 × 24 (1)
- 360 m (1)

ECF from (a) and (b)

Worked solution

(a) Acceleration = change in velocity / time, so change in velocity = acceleration × time = 0.60 m / s^2 × 40 s = 24 m / s. The train starts from rest, so its top speed is 24 m / s.

(b) While stopping, the velocity changes by 24 m / s at 0.80 m / s every second. Time = 24 / 0.80 = 30 s. Check: 0.80 × 30 = 24 m / s.

(c) While decelerating steadily, the speed–time graph is a straight line from 24 m / s down to zero over 30 s. The distance is the area of this triangle: ½ × 30 × 24 = 360 m. Check: the average speed is 12 m / s, and 12 × 30 = 360 m.

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29 · Part 1 Core Motion [6]

Mark scheme

(a) **[2]**

- speed = distance / time = 120 / 20.0 OR 30 / 5.0 (1)
- 6.0 m / s (1)

(b) **[1]**

- (130 − 50) / 20.0 = 4.0 m / s (1)

(c) **[2]**

- at t = 20.0 s P is 10 m behind Q and gains 2.0 m every second / 6.0t = 50 + 4.0t (1)
- t = 25 s (1)

ALLOW answers from extending the table to show both at 150 m at 25 s

ECF from (a) and (b)

(d) **[1]**

- no, because velocity is speed in a given direction and R moves in a different (opposite) direction (1)

Worked solution

(a) P travels 30 m in every 5.0 s, so its speed is 30 / 5.0 = 6.0 m / s. The same answer comes from 120 m / 20.0 s.

(b) Q moves from 50 m to 130 m, a distance of 80 m, in 20.0 s. Its speed is 80 / 20.0 = 4.0 m / s. Q started 50 m ahead of the signpost, so you must use the change in distance, not the distance itself.

(c) At t = 20.0 s, P is at 120 m and Q is at 130 m, so P is 10 m behind. P is faster by 6.0 − 4.0 = 2.0 m / s, so it closes the gap in 10 / 2.0 = 5.0 s. P catches Q at 20.0 + 5.0 = 25 s. Check: at 25 s, P is at 6.0 × 25 = 150 m and Q is at 50 + 4.0 × 25 = 150 m.

(d) Velocity is speed in a given direction. R has the same speed as P but moves the opposite way, so their velocities are different.

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30 · Part 1 Extended Motion [5]

Mark scheme

(a) **[2]**

- changing / decreasing acceleration (1)
- the speed increases by different amounts in each 0.10 s (0.42, 0.26, 0.15 m / s), and these increases get smaller (1)

(b) **[1]**

- 0.90 m / s (1)

(c) **[2]**

- as the speed increases, the resistance of the syrup (drag) on the ball increases (1)
- when the upward force(s) become equal to the weight, the resultant force is zero, so the ball stops accelerating and moves at a constant speed (1)

ALLOW upward forces described as drag / friction / viscous force, with or without upthrust

Worked solution

(a) In the first three intervals the speed rises by 0.42, 0.26 and 0.15 m / s. The rises are not equal and they get smaller, so the acceleration is changing: it decreases.

(b) From 0.40 s onwards the speed stays at 0.90 m / s, so this is the terminal velocity.

(c) At first the ball is moving slowly, so the resistive force from the syrup is small and the ball accelerates downwards because of its weight. The faster it moves, the larger the resistive force becomes. Eventually the upward forces balance the weight. The resultant force is then zero, so there is no acceleration, and the ball keeps falling at a constant speed.

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1 · Part 2 Core Mass and weight [1]

Answer: B

Worked solution

Moving an object does not add or remove any matter, so its mass stays the same everywhere. Weight = mass × gravitational field strength, so a smaller field strength on Mars gives a smaller weight for the same mass.

2 · Part 2 Core Mass and weight [1]

Answer: D

Worked solution

Gravitational field strength is force per unit mass, so g = W / m. Substituting: g = 1.5 / 2.5 = 0.60 N / kg. Check: 2.5 kg × 0.60 N / kg = 1.5 N, which matches the weight given.

3 · Part 2 Core Mass and weight [1]

Answer: A

Worked solution

A beam balance compares the downward pull of gravity on each pan. When the beam is level, the weights of P and Q are equal. Since W = mg and both bags experience the same g, equal weights mean equal masses too. Two bags can have equal masses and very different volumes, so the balance tells you nothing about volume.

4 · Part 2 Core Density [1]

Answer: C

Worked solution

Density is mass per unit volume: ρ = m / V = 540 / 60 = 9.0 g / cm^3. Check: 9.0 g / cm^3 × 60 cm^3 = 540 g. Because the mass is in g and the volume is in cm^3, the unit of the answer is g / cm^3.

5 · Part 2 Core Density [1]

Answer: A

Worked solution

Work out each density with ρ = m / V:

- A: 120 / 160 = 0.75 g / cm^3
- B: 40 / 25 = 1.6 g / cm^3
- C: 90 / 75 = 1.2 g / cm^3
- D: 260 / 200 = 1.3 g / cm^3

An object floats in water if its density is less than 1.0 g / cm^3. Only A does. A quick check: an object floats only if its mass in g is smaller than its volume in cm^3, which is true only for A.

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6 · Part 2 Extended Density [1]

Answer: B

Worked solution

First find the density of Y: ρ = m / V = 63 / 50 = 1.26 g / cm^3. Compare all three: X 0.92, Z 1.00, Y 1.26 g / cm^3. A less dense liquid floats on a denser liquid that it does not mix with, so the least dense liquid, X, ends up on top and the densest, Y, ends up at the bottom. The order in which they were poured does not matter once they have settled.

7 · Part 2 Core Effects of forces [1]

Answer: D

Worked solution

Take east as positive. Resultant = 120 + 45 − 30 = 135 N. The answer is positive, so the resultant force acts towards the east. Check: the eastward forces (165 N) are larger than the westward force (30 N), so the resultant must point east.

8 · Part 2 Core Effects of forces [1]

Answer: C

Worked solution

An object only moves in a straight line at constant speed if there is no resultant force on it. The cyclist is not moving in a straight line: the direction of motion changes all the time. So a resultant force must act on the cyclist. Here that force changes only the direction, not the speed.

9 · Part 2 Core Effects of forces [1]

Answer: B

Worked solution

Solid friction is the force between two surfaces that may impede motion and produce heating, so the surfaces get warmer, not colder. Drag is friction on an object moving through a liquid or a gas: a boat moving through water and a ball moving through air both experience drag.

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10 · Part 2 Extended Effects of forces [1]

Answer: A

Worked solution

The spring constant is force per unit extension, so you need the extension, not the length. Extension = 16.0 − 12.0 = 4.0 cm. Then k = F / x = 6.0 N / 4.0 cm = 1.5 N / cm. Check: a 6.0 N load on a spring with k = 1.5 N / cm gives x = 6.0 / 1.5 = 4.0 cm, which matches.

11 · Part 2 Extended Effects of forces [1]

Answer: D

Worked solution

In F = ma, F is the resultant force. The friction acts against the forward force, so the resultant is 8.0 − 3.0 = 5.0 N. Rearranging F = ma gives a = F / m = 5.0 / 2.5 = 2.0 m / s^2. Check: 2.5 kg × 2.0 m / s^2 = 5.0 N, which is the resultant force.

12 · Part 2 Extended Effects of forces [1]

Answer: C

Worked solution

Use the three rules for circular motion with a force perpendicular to the motion:

- with mass and radius constant, a larger force goes with a larger speed, so a lower speed needs less force (B);
- with mass and speed constant, a larger force makes the radius smaller, so a larger radius needs less force (D);
- with speed and radius constant, a larger mass needs a larger force (C), and a smaller mass needs less force (A).

Only C needs a larger friction force.

13 · Part 2 Core Turning effect of forces [1]

Answer: A

Worked solution

The pivot is the centre of the nut. Moment = F × d = 40 × 0.25 = 10 Nm. Check: 10 Nm / 0.25 m = 40 N, the force given. If the distance had been used in cm, the answer would be 1000 N cm, which is a different unit and is the same as 10 Nm.

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14 · Part 2 Core Turning effect of forces [1]

Answer: B

Worked solution

For balance, the anticlockwise moment equals the clockwise moment about the pivot. Moment of first child = 300 N × 1.6 m = 480 Nm. The second child must give 480 Nm on the other side: d = 480 / 400 = 1.2 m. The heavier child sits closer to the pivot, which makes sense. Check: 400 × 1.2 = 480 Nm.

15 · Part 2 Extended Turning effect of forces [1]

Answer: C

Worked solution

Take moments about the pivot. Left side total = (20 × 0.40) + (15 × 0.20) = 11.0 Nm. Right side so far = 30 × 0.30 = 9.0 Nm. The extra force must provide 11.0 − 9.0 = 2.0 Nm at 0.50 m, so F = 2.0 / 0.50 = 4.0 N. Check: 9.0 + 4.0 × 0.50 = 11.0 Nm, equal to the left side.

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16 · Part 2 Core Mass and weight [5]

Mark scheme

(a) **[1]**

- a measure of the quantity of matter in an object (at rest relative to the observer) (1)

REJECT weight / force

(b) **[1]**

- (weight is) a gravitational force on an object that has mass (1)

ALLOW the force of gravity on an object

(c) **[1]**

- force per unit mass (1)

ALLOW weight per unit mass / g = W / m with symbols explained

REJECT force × mass

(d) **[2]**

- W = mg or W = 25 × 9.8 (1)
- 250 N (1)

ALLOW 245 N

Worked solution

(a) Mass tells you how much matter the crate contains. It is measured in kg and does not depend on where the crate is.

(b) Weight is a force: the pull of gravity on the crate because it has mass. It is measured in N.

(c) Gravitational field strength is the force on each kilogram of mass, so g = W / m.

(d) Rearranging g = W / m gives W = mg = 25 kg × 9.8 N / kg = 245 N, which is 250 N to two significant figures. Check: 245 / 25 = 9.8 N / kg.

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17 · Part 2 Core Mass and weight [7]

Mark scheme

(a) **[2]**

- W = mg or W = 1200 × 3.7 (1)
- 4400 N (1)

ALLOW 4440 N

(b) **[3]**

- g = W / m or g = 1680 / 1200 (1)
- 1.4 N / kg (1)
- Titan (1)

ECF identification of the body that matches the candidate's calculated value

(c) **[1]**

- 3.7 m / s^2 (1)

REJECT 3.7 N / kg (wrong unit for an acceleration)

(d) **[1]**

- 1200 kg (1)

Worked solution

(a) Weight = mass × gravitational field strength = 1200 × 3.7 = 4440 N, which is 4400 N to two significant figures.

(b) g = W / m = 1680 / 1200 = 1.4 N / kg. This matches Titan in the table. Check: 1200 × 1.4 = 1680 N.

(c) Gravitational field strength is equivalent to the acceleration of free fall, so on Mars a falling object accelerates at 3.7 m / s^2 (ignoring air resistance).

(d) Mass is the quantity of matter in the lander. It is the same everywhere, so it is still 1200 kg on Titan. Only the weight changes from place to place.

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18 · Part 2 Extended Mass and weight [6]

Mark scheme

(a) **[1]**

- the (Earth's) gravitational field exerts a force on the sample because the sample has mass (1)

> **ALLOW** weight is the effect of a gravitational field on a mass

(b) **[2]**

- Earth: 7.4 N (1)

> **ALLOW** 7.35 N

- Moon: 1.2 N (1)

> **ECF** Moon reading consistent with a correct method but a wrong Earth value, e.g. if the wrong mass was used in both

(c) **[3]**

- the beam balance compares the weight of the sample with the weight of the standard masses (1)
- both are in the same gravitational field, so their weights both decrease by the same factor and stay equal (so the same masses balance) (1)
- the newtonmeter measures the weight (force) of the sample, which depends on the gravitational field strength, so it reads less on the Moon (1)

> **ALLOW** equivalent statements, e.g. "the masses do not change, so a balance comparing them still balances"

Worked solution

(a) Any object with mass in a gravitational field has a force exerted on it by the field. This force is its weight.

(b) Weight = mg. On Earth: 0.75 × 9.8 = 7.35 N, which is 7.4 N to two significant figures. On the Moon: 0.75 × 1.6 = 1.2 N.

(c) A beam balance is level when the weights on the two pans are equal. On the Moon the sample and the standard masses are in the same weaker field, so both weights are smaller by the same factor (1.6 / 9.8 of their Earth values) and they are still equal. So the beam balance really compares masses, and the masses have not changed. The newtonmeter measures the force on the sample directly. That force depends on g, so the reading falls from 7.4 N to 1.2 N.

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19 · Part 2 Core Density [7]

Mark scheme

(a) **[1]**

- 24.0 cm^3 (1)

ALLOW 24 cm^3

(b) **[2]**

- ρ = m / V or ρ = 64.8 / 24.0 (1)
- 2.70 g / cm^3 (1)

ALLOW 2.7 g / cm^3; 2700 kg / m^3

ECF from the volume in (a)

(c) **[3]**

- measure the mass of the empty measuring cylinder on the balance (1)
- pour some of the liquid into the cylinder, read its volume, and measure the mass of the cylinder and liquid; mass of liquid = difference between the two masses (1)
- density = mass of liquid / volume of liquid (1)

ALLOW zeroing (taring) the balance with the empty cylinder on it, for the first two marking points

(d) **[1]**

- any one from: read the scale with the eye level with the liquid surface; read the bottom of the meniscus; make sure there are no air bubbles stuck to the stone; make sure the stone is completely covered by the water (1)

Worked solution

(a) The stone pushes aside its own volume of water, so its volume is the rise in the reading: 66.0 − 42.0 = 24.0 cm^3.

(b) ρ = m / V = 64.8 / 24.0 = 2.70 g / cm^3. Check: 2.70 × 24.0 = 64.8 g.

(c) Find the mass of the empty measuring cylinder. Pour in some liquid and read its volume from the scale. Find the mass of the cylinder with the liquid in it. Subtract to get the mass of the liquid alone, then divide this mass by the volume.

(d) Reading at eye level, at the bottom of the meniscus, avoids parallax errors and gives an accurate volume.

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20 · Part 2 Core Density [7]

Mark scheme

(a) **[1]**

- 40 cm^3 (1)

(b) **[2]**

- $\rho = m / V$ or $\rho = 32 / 40$ (1)
- 0.80 g / cm^3 (1)

ECF from the volume in (a)

(c) **[2]**

- floats in water, because its density (0.80 g / cm^3) is less than the density of water (1)
- sinks in liquid L, because its density is greater than the density of L (1)

ECF from the density in (b)

(d) **[2]**

- total mass = 47 g and total volume = 42 cm^3 (1)
- density = 1.1 g / cm^3, which is greater than the density of water, so it sinks (1)

ALLOW 1.12 g / cm^3

ECF from (a)

REJECT a decision with no supporting density (or equivalent mass and volume comparison)

Worked solution

(a) Volume of a rectangular block = length × width × height = 5.0 × 4.0 × 2.0 = 40 cm^3.

(b) $\rho = m / V = 32 / 40 = 0.80$ g / cm^3.

(c) An object floats in a liquid if its density is less than the liquid's density. 0.80 < 1.0, so the block floats in water. 0.80 > 0.70, so the block sinks in liquid L.

(d) Treat the block and metal as one object. Total mass = 32 + 15 = 47 g. Total volume = 40 + 2.0 = 42 cm^3. Average density = 47 / 42 = 1.12 g / cm^3, which is 1.1 g / cm^3. This is greater than 1.0 g / cm^3, so the combined object sinks. Another way to check: the combined object would need a volume larger than 47 cm^3 to float in water, but it has only 42 cm^3.

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21 · Part 2 Core Effects of forces [5]

Mark scheme

(a) **[2]**

- 2400 − 900 − 300 = 1200 N (1)
- forwards / in the direction of the driving force (1)

(b) **[1]**

- the speed increases (1)

ALLOW the van accelerates

ECF from the direction in (a)

(c) **[2]**

- at constant speed in a straight line there is no resultant force / the forces are balanced (1)
- air resistance = 2000 − 300 = 1700 N (1)

Worked solution

(a) Take forwards as positive. The two backward forces subtract from the driving force: 2400 − 900 − 300 = 1200 N. The answer is positive, so the resultant force acts forwards.

(b) A resultant force in the direction of motion makes the van speed up.

(c) An object moving in a straight line at constant speed has no resultant force on it, so the forward force equals the total backward force. 2000 = 300 + air resistance, so air resistance = 1700 N. Check: 300 + 1700 = 2000 N.

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22 · Part 2 Core Effects of forces [6]

Mark scheme

(a) **[1]**

- drag (1)

ALLOW friction (drag) from the water / water resistance

(b) **[2]**

- solid friction is the force between two surfaces that impedes motion (1)
- (solid) friction produces heating (of the surfaces), so the underside warms up (1)

ALLOW work done against friction is transferred to thermal energy

(c) **[2]**

- drag (friction from the water) acts on the kayak opposite to its direction of motion (1)
- there is now a resultant force backwards (because there is no forward force from paddling), which reduces the speed (1)

(d) **[1]**

- any one sensible suggestion, e.g. put the kayak on wheels or rollers / carry it so that it is not in contact with the sand / pull it over a smooth or wet surface (1)

Worked solution

(a) Friction on an object moving through a liquid is called drag.

(b) Solid friction is the force between two surfaces that may impede (oppose) motion. Rubbing the two surfaces against each other produces heating, so the bottom of the kayak gets warm.

(c) When paddling stops there is no forward force, but drag still acts backwards on the moving kayak. This gives a backward resultant force, and a resultant force against the motion makes the kayak slow down.

(d) Anything that reduces the rubbing between the kayak and the sand reduces the friction, for example wheels, rollers, or lifting the kayak off the ground.

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23 · Part 2 Core Effects of forces [5]

Mark scheme

(a) **[1]**

- a force can change the shape (or size) of an object (1)

(b) **[3]**

- change the speed (1)
- change the direction (of motion) (1)
- both happen to the ball (1)

(c) **[1]**

- it continues to move in a straight line at constant speed (1)

REJECT slows down / stops

Worked solution

(a) The ball is squashed, so the force from the racket changes its shape.

(b) A resultant force can change an object's velocity by changing its speed, its direction of motion, or both. The ball reverses direction, so its direction changes, and it leaves faster than it arrived, so its speed changes too.

(c) With no force acting, there is no resultant force, so the ball neither speeds up, slows down nor changes direction. It keeps moving in a straight line at a constant speed.

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24 · Part 2 Extended Effects of forces [8]

Mark scheme

(a) **[1]**

- force per unit extension (1)

ALLOW k = F / x with symbols explained

(b) **[2]**

- extension worked out from the table, e.g. 18.0 − 8.0 = 10.0 cm for 4.0 N (1)
- k = F / x = 4.0 / 10.0 = 0.40 N / cm (1)

ALLOW 40 N / m; any correct pair of load and extension in the range 0 to 4.0 N

REJECT k calculated from the length instead of the extension (e.g. 4.0 / 18.0) for the second mark

(c) **[3]**

- the limit of proportionality is the point beyond which the extension is no longer proportional to the load (1)
- up to 4.0 N, each 1.0 N of load increases the length by the same amount, 2.5 cm (or load / extension is constant) (1)
- from 4.0 N to 5.0 N the length increases by 3.1 cm, not 2.5 cm, so the limit is reached between 4.0 N and 5.0 N (1)

(d) **[2]**

- extension = 2.6 / 0.40 = 6.5 cm (1)
- length = 8.0 + 6.5 = 14.5 cm (1)

ECF from the spring constant in (b)

Worked solution

(a) The spring constant is the force needed per unit extension: k = F / x.

(b) Extension = stretched length − unloaded length. At 4.0 N: 18.0 − 8.0 = 10.0 cm. So k = 4.0 / 10.0 = 0.40 N / cm. Check with another row: at 2.0 N the extension is 5.0 cm, and 2.0 / 5.0 = 0.40 N / cm.

(c) The limit of proportionality is where the load–extension relationship stops being proportional. For each extra 1.0 N up to 4.0 N, the spring stretches a further 2.5 cm. From 4.0 N to 5.0 N it stretches 3.1 cm, and from 5.0 N to 6.0 N it stretches 3.9 cm, so it is no longer proportional. The limit is reached somewhere between 4.0 N and 5.0 N.

(d) 2.6 N is below the limit, so x = F / k = 2.6 / 0.40 = 6.5 cm. The length is the unloaded length plus the extension: 8.0 + 6.5 = 14.5 cm. Check: it lies between the lengths for 2.0 N (13.0 cm) and 3.0 N (15.5 cm).

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25 · Part 2 Extended Effects of forces [6]

Mark scheme

(a) **[1]**

- 340 − 100 = 240 N (1)

(b) **[2]**

- a = F / m or a = 240 / 80 (1)
- 3.0 m / s^2 (1)

ECF from (a)

(c) **[1]**

- forwards / the same direction as the resultant force (1)

(d) **[2]**

- 560 / 80 = 7.0 m / s^2 (1)
- backwards / opposite to the direction of motion / the same direction as the backward force (1)

ALLOW −7.0 m / s^2 if forwards is clearly taken as positive

Worked solution

(a) The resistive force acts against the motor force, so the resultant force = 340 − 100 = 240 N forwards.

(b) Rearranging F = ma gives a = F / m = 240 / 80 = 3.0 m / s^2. Check: 80 × 3.0 = 240 N.

(c) The acceleration is always in the same direction as the resultant force, so it is forwards.

(d) Now the resultant force is 560 N backwards. a = 560 / 80 = 7.0 m / s^2. The acceleration is in the same direction as the resultant force, which is backwards, so the scooter slows down.

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26 · Part 2 Core Effects of forces [6]

Mark scheme

(a) **[2]**

- W = mg or W = 0.50 × 9.8 (1)
- 4.9 N (1)

(b) **[2]**

- 12 − 4.9 − 2.0 = 5.1 N (1)
- upwards (1)

ECF from the weight in (a)

(c) **[2]**

- downwards (1)
- the speed decreases (1)

Worked solution

(a) Weight = mass × gravitational field strength = 0.50 × 9.8 = 4.9 N.

(b) Take upwards as positive. The weight acts downwards and, because the rocket moves upwards, the air resistance also acts downwards. Resultant = 12 − 4.9 − 2.0 = 5.1 N, which is positive, so it acts upwards.

(c) With the motor off, the only forces are the weight (downwards) and the air resistance (opposite to the upward motion, so also downwards). The resultant force is therefore downwards, opposite to the motion, so the rocket slows down.

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27 · Part 2 Core Turning effect of forces [6]

Mark scheme

(a) **[2]**

- the turning effect of a force (1)
- any sensible everyday example other than a door, e.g. using a spanner, a seesaw, a wheelbarrow, a bottle opener, turning a steering wheel, a crowbar (1)

(b) **[1]**

- moment = force × perpendicular distance from the pivot (1)

(c) **[2]**

- moment = 15 × 0.80 (1)
- 12 Nm (1)

(d) **[1]**

- a larger distance from the hinges gives a larger moment for the same force / so a smaller force is needed to open the door (1)

Worked solution

(a) The moment of a force measures how strongly the force turns an object about a pivot. Tightening a nut with a spanner is an everyday example.

(b) Moment = force × perpendicular distance from the pivot to the line of the force.

(c) The force is at right angles to the door, so the perpendicular distance is 0.80 m. Moment = 15 × 0.80 = 12 Nm.

(d) For the same push, a larger distance gives a larger turning effect. So putting the handle far from the hinges means a smaller force is needed to open the door.

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28 · Part 2 Core Turning effect of forces [6]

Mark scheme

(a) **[2]**

- moment = 150 × 1.4 (1)
- 210 Nm (1)

(b) **[2]**

- F × 1.5 = 210 (principle of moments) (1)
- F = 140 N (1)

ECF from (a)

(c) **[2]**

- there is no resultant force (on the object) (1)
- there is no resultant moment (on the object) (1)

Worked solution

(a) Moment = force × perpendicular distance from the pivot = 150 × 1.4 = 210 Nm.

(b) For balance, the clockwise moment of the worker's push equals the anticlockwise moment of the box. F × 1.5 = 210, so F = 210 / 1.5 = 140 N. Check: 140 × 1.5 = 210 Nm.

(c) An object is in equilibrium when there is no resultant force and no resultant moment acting on it. Both are needed: no resultant force stops it moving off, and no resultant moment stops it turning.

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29 · Part 2 Extended Turning effect of forces [7]

Mark scheme

(a) **[2]**

- (150 × 0.40) + (500 × 0.36) or 60 + 180 (1)
- 240 Nm (1)

(b) **[2]**

- F × 1.20 = 240 (1)
- F = 200 N (1)

ECF from (a)

(c) **[2]**

- no resultant force, so upward forces = downward forces: force from ground + F = 150 + 500 (1)
- 450 N (1)

ECF from F in (b)

(d) **[1]**

- place the sand closer to the wheel (axle) / further from the handles (1)

Worked solution

(a) Each weight gives a moment about the axle: 150 × 0.40 = 60 Nm and 500 × 0.36 = 180 Nm. Both turn the same way, so the total is 60 + 180 = 240 Nm.

(b) The ground's force acts directly below the axle, so it has no moment about the axle. For balance, the moment of F must equal 240 Nm in the opposite direction: F × 1.20 = 240, so F = 200 N.

(c) The wheelbarrow is held still, so there is no resultant force. Total upward force = total downward force: ground force + 200 = 150 + 500 = 650 N, so the ground force = 450 N.

(d) Moving the sand closer to the axle reduces its perpendicular distance from the pivot, so its moment is smaller and a smaller force F is needed. Check: if the sand acted 0.24 m from the axle, the total moment would be 60 + 120 = 180 Nm and F would be 150 N.

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30 · Part 2 Extended Turning effect of forces [6]

Mark scheme

(a) **[4]**

- balance the metre rule (horizontally) on the pivot with no weights on it (1)
- hang weights on both sides of the pivot and adjust their positions until the rule balances horizontally (1)
- measure each weight (force) and its distance from the pivot (1)
- calculate the total clockwise moment and the total anticlockwise moment and show that they are equal; repeat with different weights or positions (including more than one weight on a side) (1)

(b) **[2]**

- anticlockwise moment = 2.0 × 0.300 = 0.60 Nm and clockwise moment = 1.5 × 0.400 = 0.60 Nm (1)
- the moments are equal (and opposite), so there is no resultant moment (1)

ALLOW the rule's own weight has no moment because it is balanced at its centre

Worked solution

(a) First balance the rule on the pivot by itself, so that the rule's own weight does not turn it. Hang known weights on each side and slide them until the rule is level again; the rule is now in equilibrium. Measure the distance of each weight from the pivot. Multiply each weight by its distance to find its moment. Add the moments that turn the rule clockwise, and add those that turn it anticlockwise. The two totals are equal, so there is no resultant moment. Repeat with other weights and positions to show that this is always true when the rule balances.

(b) Left (anticlockwise): 2.0 × 0.300 = 0.60 Nm. Right (clockwise): 1.5 × 0.400 = 0.60 Nm. The moments are equal in size and opposite in direction, so the resultant moment is zero, which is what the balanced rule shows.

1 · Part 3 Core Centre of gravity [1]

Answer: D

Explanation

The centre of gravity is the point through which all of the weight of an object appears to act. It is only at the geometric centre for a regular, uniform object (A); it is not defined as the lowest point (B); and it has nothing to do with where the density is greatest (C).

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2 · Part 3 Core Centre of gravity [1]

Answer: A

Explanation

A wider base and a heavier base both make the lamp more stable: the centre of gravity is lower, and the lamp must tilt further before the centre of gravity moves outside the base. A narrower base (B) makes it less stable. A heavy weight near the top (C) or a taller stem (D) raises the centre of gravity and makes the lamp less stable.

3 · Part 3 Core Centre of gravity [6]

Mark scheme

(a) **[1]**

- the point through which all of the weight (of the object) appears to act (1)

> **ALLOW** the point where all the weight seems to act
>
> **REJECT** the centre of the object / where all the mass is

(b) **[4]**

- make a hole near the edge of the card and hang the card from the pin so that it can swing freely (1)
- hang the plumb line from the same pin and wait until the card and the plumb line are still (1)
- draw a line on the card along the plumb line (1)
- repeat with a second hole at a different place near the edge; the centre of gravity is where the two lines cross (1)

> **ALLOW** marks for an equivalent clear sequence in any order of wording

(c) **[1]**

- the third line passes through the same point where the first two lines cross (1)

Worked solution

(a) The centre of gravity is the single point through which the whole weight of an object appears to act.

(b) When the card hangs freely from a pin and comes to rest, its centre of gravity is directly below the pin. A plumb line hung from the pin shows the vertical line through the pin, so the centre of gravity lies somewhere on the line you draw along the thread. Hanging the card from a second hole gives a second line; the centre of gravity must be on both lines, so it is where they cross.

(c) If the method is accurate, a line from any other hole must also pass through that same crossing point, so a third line through the same point confirms the result.

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4 · Part 3 Extended Momentum [1]

Answer: C

Explanation

Momentum = mass × velocity = 85 × 6.0 = 510 kg m / s (C). A divides mass by velocity; B adds them; D is the kinetic energy ½mv^2 in joules, not momentum.

5 · Part 3 Extended Momentum [1]

Answer: B

Worked solution

Total momentum before = 2.0 × 3.0 + 1.0 × 0 = 6.0 kg m / s. Total momentum after = (2.0 + 1.0) × v. So 3.0 v = 6.0 and v = 2.0 m / s.

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6 · Part 3 Extended Momentum [6]

Mark scheme

(a) **[3]**

- taking one direction as positive, the velocities are +8.0 m / s and −6.0 m / s, or change in velocity = 14 m / s (1)
- $\Delta p = m \times \Delta v = 0.27 \times 14$ (1)
- 3.8 kg m / s (1)

ALLOW 3.78 kg m / s; ALLOW: −3.8 kg m / s; ALLOW: N s as the unit

REJECT 0.54 kg m / s (from 8.0 − 6.0) for the final mark, but award the second mark if $0.27 \times \Delta v$ is shown with their Δv

(b) **[1]**

- 3.8 N s (or kg m / s), the same value as the change in momentum (1)

ECF their (a) with a correct unit

(c) **[2]**

- $F = \Delta p / \Delta t = 3.78 / 0.090$ (1)
- 42 N (1)

ECF their (a) / 0.090

ALLOW 42.2 N or 42.0 N

Worked solution

Velocity is a vector, so direction matters. Take "towards the wall" as positive: initial velocity = +8.0 m / s, final velocity = −6.0 m / s. The change in velocity has size 8.0 + 6.0 = 14 m / s (not 2.0 m / s – the ball does not just slow down, it reverses).

(a) $\Delta p = 0.27 \times 14 = 3.78$ kg m / s, which is 3.8 kg m / s to 2 significant figures.

(b) Impulse = change in momentum, so the impulse on the ball is 3.8 N s (1 N s = 1 kg m / s).

(c) Resultant force = change in momentum per unit time: $F = 3.78 / 0.090 = 42$ N.

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7 · Part 3 Extended Momentum [6]

Mark scheme

(a) **[1]**

- momentum = mass × velocity (1)

ALLOW p = mv with symbols explained

(b) **[1]**

- zero (1)

(c) **[3]**

- momentum of P = 64 × 1.5 = 96 kg m / s (1)
- momentum of Q has the same size (so that total momentum stays zero) / 48 × v = 96 (1)
- 2.0 m / s (1)

ECF their momentum of P / 48

(d) **[1]**

- in the opposite direction to P (1)

Worked solution

(a) Momentum is mass multiplied by velocity.

(b) Both skaters are at rest, so each has zero momentum and the total is zero.

(c) Momentum is conserved, so the total momentum after the push is still zero. P has momentum 64 × 1.5 = 96 kg m / s in one direction, so Q must have 96 kg m / s in the other direction. 48 × v = 96, so v = 96 / 48 = 2.0 m / s. The lighter skater moves faster.

(d) For the two momenta to add to zero they must be in opposite directions, so Q moves the opposite way to P.

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8 · Part 3 Extended Momentum [7]

Mark scheme

(a) **[1]**

- force × time for which the force acts (1)

ALLOW impulse = FΔt

(b) **[2]**

- Δp = 80 × 15 (1)
- 1200 kg m / s (1)

ALLOW −1200 kg m / s; ALLOW: N s

(c) **[2]**

- F = Δp / Δt = 1200 / 0.15 (1)
- 8000 N (1)

ECF their (b) / 0.15

(d) **[2]**

- the change in momentum of the dummy is the same (it starts at the same speed and stops) (1)
- the time taken is shorter, so the rate of change of momentum is greater, so the force is larger (1)

ALLOW F = Δp / Δt with the same Δp and smaller Δt gives larger F

Worked solution

(a) Impulse is the force multiplied by the time for which the force acts.

(b) The dummy goes from 15 m / s to rest, so Δp = 80 × 15 − 0 = 1200 kg m / s.

(c) Resultant force = change in momentum ÷ time = 1200 / 0.15 = 8000 N.

(d) In both tests the dummy loses the same momentum (1200 kg m / s). Without the airbag this happens in a much shorter time. Since F = Δp / Δt, the same Δp in a smaller Δt needs a larger force. An airbag works by making the stopping time longer, which reduces the force.

9 · Part 3 Core Energy [1]

Answer: D

Explanation

Kinetic, chemical and elastic (strain) are all energy stores (D). Heating (A), sound and light (B), and electric current (C) are ways of transferring energy between stores, not stores.

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10 · Part 3 Core Energy [1]

Answer: A

Explanation

Energy is transferred from the chemical store of the battery to the kinetic store of the spinning blades, by an electric current doing electrical work (A). B is backwards; C names the wrong final store and the wrong transfer; D names the wrong starting store and an incorrect transfer by light.

11 · Part 3 Extended Energy [1]

Answer: B

Worked solution

$\Delta Ep = 0.50 \times 9.8 \times 3.2 = 15.68$ J. This all becomes kinetic energy: $15.68 = ½ \times 0.50 \times v^2$, so $v^2 = 15.68 / 0.25 = 62.72$ and $v = 7.9$ m / s. The mass cancels, so any mass gives the same speed.

12 · Part 3 Core Energy [4]

Mark scheme

(a) **[1]**

- energy cannot be created or destroyed (it can only be transferred from one store to another) (1)

ALLOW total energy (of a closed system) stays the same

(b) **[2]**

- 900 − 135 − 45 (1)
- 720 J (1)

(c) **[1]**

- internal (thermal) store (of the television and the surroundings) (1)

ALLOW thermal store

REJECT heat store / heating (heating is a transfer, not a store)

Worked solution

(a) Energy is never created or destroyed; the total stays the same, it just moves between stores.

(b) All 900 J must go somewhere. Light and sound account for 135 + 45 = 180 J, so the rest, 900 − 180 = 720 J, is transferred by heating.

(c) Heating raises the temperature of the television and the air around it, so the energy ends up in their internal (thermal) store.

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13 · Part 3 Extended Energy [7]

Mark scheme

(a) **[2]**

- $\Delta Ep = mg\Delta h = 50 \times 9.8 \times 2.5$ (1)
- 1200 J (1)

ALLOW 1225 J or 1230 J

(b) **[3]**

- kinetic energy at the bottom = decrease in gravitational potential energy / $\frac{1}{2}mv^2 = 1225$ (1)
- $v^2 = 2 \times 1225 / 50 = 49$ (1)
- 7.0 m / s (1)

ECF from their (a)

ALLOW 6.9 m / s if 1200 J is used

(c) **[2]**

- kinetic energy at the bottom = $\frac{1}{2} \times 50 \times 5.0^2 = 625$ J (1)
- energy transferred to other stores = 1225 − 625 = 600 J (1)

ECF their (a) − 625

ALLOW 575 J or 580 J if 1200 J is used

Worked solution

(a) $\Delta Ep = mg\Delta h = 50 \times 9.8 \times 2.5 = 1225$ J, which is 1200 J to 2 significant figures.

(b) With no energy transferred elsewhere, all 1225 J goes into the kinetic store: $\frac{1}{2} \times 50 \times v^2 = 1225$, so $v^2 = 1225 / 25 = 49$ and $v = 7.0$ m / s.

(c) At 5.0 m / s the kinetic energy is only $\frac{1}{2} \times 50 \times 25 = 625$ J. The difference, 1225 − 625 = 600 J, has been transferred to other stores (mainly the internal store of the wheels, ramp and air) by friction and air resistance.

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14 · Part 3 Extended Energy [6]

Mark scheme

(a) **[2]**

- energy wasted by the engine = 1400 kJ (1)
- useful energy output of the winch and cable = 480 kJ (1)

(b) **[1]**

- internal (thermal) store of the surroundings (1)

ALLOW thermal store of the air / surroundings

(c) **[2]**

- efficiency = useful energy output / total energy input = 480 / 2000 (1)
- 0.24 or 24% (1)

ECF their useful output from (a) / 2000

REJECT 480 / 600 or 600 / 2000 for the second mark

(d) **[1]**

- lubricate / oil the winch (to reduce friction) (1)

ALLOW any sensible way to reduce friction, e.g. use smoother bearings

Worked solution

(a) Energy is conserved at each stage. Engine: 2000 − 600 = 1400 kJ wasted. Winch and cable: 600 − 120 = 480 kJ useful output.

(b) The wasted energy heats the engine, winch and cable, and that energy spreads into the surroundings, so it ends up in the internal (thermal) store of the surroundings.

(c) For the whole crane, the input is the 2000 kJ from the fuel and the final useful output is the 480 kJ in the crate's gravitational potential store. Efficiency = 480 / 2000 = 0.24, or 24%. Note that you must not use the input or output of only one stage.

(d) Most energy wasted in a winch is due to friction, so oiling moving parts reduces it.

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15 · Part 3 Core Energy [4]

Mark scheme

(a) **[1]**

- elastic (strain) store (of the band) (1)

(b) **[2]**

- energy is transferred from the elastic store of the band to the kinetic store of the pellet (by the force of the band / mechanical work) (1)
- as the pellet rises, energy is transferred from its kinetic store to its gravitational potential store (1)

> **ALLOW** some energy to the internal (thermal) store of the air because of air resistance, but this is not needed for the marks

(c) **[1]**

- in the internal (thermal) store of the surroundings / floor and air (1)

> **REJECT** in the gravitational potential store / the energy has been used up

Worked solution

(a) Before release, the stretched band stores energy in its elastic (strain) store.

(b) When released, the band pushes the pellet, doing work on it, so energy moves from the elastic store to the pellet's kinetic store. As the pellet slows down while rising, energy moves from its kinetic store to its gravitational potential store. At the highest point the pellet is momentarily still.

(c) When it lands and stops, the energy has not disappeared. It has been transferred, by air resistance and the impact (including sound), to the internal (thermal) store of the floor and the surrounding air.

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16 · Part 3 Core Work [1]

Answer: D

Explanation

Work done = force × distance moved in the direction of the force = 35 × 12 = 420 J (D). A is 12 / 35, B is 35 / 12 and C adds the force and the distance.

17 · Part 3 Core Work [5]

Mark scheme

(a) **[1]**

- the work done is equal to the energy transferred (to the lift) (1)

> **ALLOW** work done = energy transferred to the gravitational potential store of the lift

(b) **[2]**

- W = Fd = 6400 × 15 (1)
- 96 000 J (1)

> **ALLOW** 96 kJ

(c) **[2]**

- d = W / F = 48 000 / 6400 (1)
- 7.5 m (1)

Worked solution

(a) Mechanical work done is the same as the energy transferred. Here the cable does work on the lift and the energy goes into the lift's gravitational potential store.

(b) At constant speed the cable pulls up with a force equal to the weight, 6400 N. W = Fd = 6400 × 15 = 96 000 J.

(c) Rearranging W = Fd gives d = W / F = 48 000 / 6400 = 7.5 m.

18 · Part 3 Core Energy resources [1]

Answer: C

Explanation

Burning gas heats water in the boiler to make steam; the steam turns the turbine; the turbine turns the generator, which produces electrical power (C). The other orders put the parts in an impossible sequence.

19 · Part 3 Core Energy resources [1]

Answer: A

Explanation

Wind is renewable, but it is not reliable because the wind does not always blow (A). Geothermal (B) is renewable and reliable. Natural gas (C) and nuclear fuel (D) are non-renewable.

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20 · Part 3 Extended Energy resources [1]

Answer: D

Explanation

Tidal energy comes mainly from the gravitational pull of the Moon (and Sun) on the oceans, not from the Sun's radiation (D). Coal formed from plants that used sunlight (A); hydroelectric depends on water evaporated by the Sun and falling as rain (B); wind is caused by the Sun heating the Earth unevenly (C).

21 · Part 3 Extended Energy resources [1]

Answer: B

Worked solution

Efficiency must compare like with like: power with power, or energy with energy. Useful power = 4.5 kJ per minute = 4500 J / 60 s = 75 W. Efficiency = 75 / 300 = 0.25 = 25%. (Alternatively, total energy input per minute = 300 × 60 = 18 000 J, and 4500 / 18 000 = 0.25.)

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22 · Part 3 Core Energy resources [6]

Mark scheme

(a) **[3]**

- water stored behind the dam has energy in its gravitational potential store (1)
- water flows down (through pipes) and its kinetic energy increases (1)
- the moving water turns a turbine, which turns (drives) a generator (1)

ALLOW water falls through the dam to turn turbines connected to generators for the second and third marks

(b) **[2]**

- advantage: any one from: renewable / does not produce carbon dioxide (or greenhouse gases) when operating / reliable or available when needed / no fuel to buy (1)
- disadvantage: any one from: floods land or destroys habitats when the lake is made / can only be built in a few suitable places (such as mountain valleys) / expensive to build (1)

(c) **[1]**

- any one from: wind turbines are not reliable (the wind does not always blow) / each turbine produces only a small amount of power so many are needed / noise or visual pollution (1)

REJECT wind is non-renewable

Worked solution

(a) Water held high up behind the dam stores energy in its gravitational potential store. When it is let out, it flows down through pipes and gains kinetic energy. The moving water spins a turbine, and the turbine turns a generator, which produces electrical power.

(b) Hydroelectric power is renewable and releases no carbon dioxide when running, unlike burning gas, and it can be switched on quickly when needed. However, building the dam floods a large area, which destroys habitats, and it needs a suitable river valley.

(c) The wind is not always blowing, so wind turbines are less reliable than water stored behind a dam, and a single turbine produces much less power.

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23 · Part 3 Core Energy resources [5]

Mark scheme

(a) **[3]**

- water is pumped down to hot rocks underground (1)
- the water is heated and turns into steam (which rises to the surface) (1)
- the steam turns a turbine, which turns (drives) a generator (1)

ALLOW hot water or steam from underground used directly for the second mark

(b) **[1]**

- hot rocks are only close enough to the surface in a few places (such as volcanic regions) / geothermal resources are not available everywhere (1)

(c) **[1]**

- solar panels heat water but solar cells generate electrical power (1)

ALLOW solar panels use infrared from the Sun but solar cells use light

REJECT answers that just say one is bigger or more efficient

Worked solution

(a) Deep underground the rocks are very hot. Cold water is pumped down to them, is heated, and returns as steam. The steam drives a turbine, which turns a generator to produce electrical power.

(b) Geothermal power needs hot rocks near the surface, which only happens in certain areas, for example where there are volcanoes. So it is limited by availability.

(c) A solar panel absorbs infrared (and other radiation) from the Sun to heat water. A solar cell turns light directly into electrical power.

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24 · Part 3 Extended Energy resources [6]

Mark scheme

(a) **[1]**

- nuclear fusion (1)

REJECT fission / burning

(b) **[1]**

- no (it is still being researched) (1)

(c)(i) **[2]**

- efficiency = useful power output / total power input = 990 / 3000 (1)
- 0.33 or 33% (1)

(c)(ii) **[1]**

- 3000 − 990 = 2010 MW (1)

ALLOW 2000 MW

(c)(iii) **[1]**

- to the internal (thermal) store of the surroundings, e.g. by heating cooling water, the air or a river (1)

ALLOW lost by heating the surroundings

Worked solution

(a) The Sun releases energy by nuclear fusion: light nuclei join together.

(b) Fusion is being researched but is not yet used to generate electrical power on a large scale.

(c)(i) Efficiency = useful power output / total power input = 990 / 3000 = 0.33, which is 33%.

(ii) Power wasted = power input − useful power output = 3000 − 990 = 2010 MW.

(iii) Most wasted energy heats the surroundings: it is carried away by cooling water and into the air, so it goes to the internal (thermal) store of the surroundings.

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25 · Part 3 Extended Energy resources [6]

Mark scheme

(a) **[1]**

- 1.6 × 1000 = 1600 W (1)

(b) **[2]**

- efficiency = useful power output / total power input = 240 / 1600 (1)
- 0.15 or 15% (1)

ECF 240 / their (a)

(c) **[2]**

- E = Pt = 240 × 2.0 × 3600 / time converted to 7200 s (1)
- 1.7×10^6 J (1)

ALLOW 1 728 000 J; ALLOW: 0.48 kW h

REJECT 480 J (time not converted) for the second mark

(d) **[1]**

- energy cannot be created, so the useful energy output cannot be more than the energy input (1)

ALLOW reference to the principle of conservation of energy

Worked solution

(a) Each square metre receives 1000 W, and there are 1.6 m^2, so the total power input is 1.6 × 1000 = 1600 W.

(b) Efficiency = useful power output / total power input = 240 / 1600 = 0.15, or 15%.

(c) Power is energy transferred per unit time, so E = P × t. Time must be in seconds: 2.0 h = 2.0 × 3600 = 7200 s. E = 240 × 7200 = 1 728 000 J, which is 1.7×10^6 J to 2 significant figures.

(d) By conservation of energy, the panel cannot output more energy than it receives. Some input is always wasted (for example heating the panel), so the efficiency is always less than 100%.

26 · Part 3 Core Power [1]

Answer: A

Explanation

Work done = 520 × 3.5 = 1820 J, and power = work done / time = 1820 / 4.0 = 455 W, which is 460 W to 2 significant figures (A). B is the work done in joules, not the power. C multiplies the weight by the time. D multiplies the work done by the time instead of dividing.

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27 · Part 3 Core Pressure [1]

Answer: B

Explanation

Pressure = force / area. The force (the weight of the rucksack) is the same, but wide straps have a larger area of contact, so the pressure is smaller (B). A and D are wrong because the weight and force do not change. C has the relationship the wrong way round.

28 · Part 3 Extended Pressure [1]

Answer: C

Worked solution

Use the change in depth, not the final depth: $\Delta h = 15$ m. $\Delta p = 1030 \times 9.8 \times 15 = 1.5 \times 10^5$ Pa.

29 · Part 3 Core Pressure [6]

Mark scheme

(a) **[1]**

- force per unit area (1)

ALLOW p = F / A with symbols explained

(b) **[3]**

- area of smallest face = 0.40 × 0.30 = 0.12 m^2 (1)
- p = F / A = 2400 / 0.12 (1)
- 20 000 N / m^2 (1)

ECF 2400 / their area

ALLOW 2.0×10^4 N / m^2; ALLOW: Pa as the unit

(c) **[2]**

- the pressure decreases (1)
- the force (weight) is the same but the area in contact with the ground is larger (1)

Worked solution

(a) Pressure is the force acting per unit area.

(b) The smallest face is made from the two shortest sides: 0.40 m × 0.30 m = 0.12 m^2. The force on the ground is the weight, 2400 N. p = F / A = 2400 / 0.12 = 20 000 N / m^2.

(c) On its largest face (0.50 m × 0.40 m = 0.20 m^2) the weight is unchanged, but it is spread over a bigger area, so the pressure is smaller (12 000 N / m^2).

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30 · Part 3 Extended Pressure [6]

Mark scheme

(a)(i) **[1]**

- pressure increases as the depth increases (1)

ALLOW pressure is proportional to depth

(a)(ii) **[1]**

- the greater the density, the greater the pressure (at the same depth) (1)

(b) **[2]**

- $\Delta p = \rho g \Delta h = 1000 \times 9.8 \times 1.8$ (1)
- 1.8×10^4 Pa (1)

ALLOW 17 640 Pa or 17 600 Pa; ALLOW: N / m^2

(c) **[2]**

- area = 2.0×10^{-4} m^2 and $F = pA = 17\,640 \times 2.0 \times 10^{-4}$ (1)
- 3.5 N (1)

ECF their (b) $\times 2.0 \times 10^{-4}$

ALLOW 3.6 N if 1.8×10^4 Pa is used

REJECT 35 000 N (area not converted from cm^2) for the second mark

Worked solution

(a) The deeper you go in a liquid, the greater the pressure, because there is more liquid above pushing down. At the same depth, a denser liquid gives a greater pressure.

(b) $\Delta p = \rho g \Delta h = 1000 \times 9.8 \times 1.8 = 17\,640$ Pa, which is 1.8×10^4 Pa to 2 significant figures.

(c) Convert the area to m^2: 1 $cm^2 = 1 \times 10^{-4}$ m^2, so 2.0 $cm^2 = 2.0 \times 10^{-4}$ m^2. Rearranging $p = F / A$ gives $F = pA = 17\,640 \times 2.0 \times 10^{-4} = 3.5$ N.

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